

Jiayong Lin

Machine Learning Engineer

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SUMMARY

Staff-level AI Engineer with 10+ years of application development experience building and deploying machine learning systems in cloud-first environments. Deep expertise in Python, PyTorch, TensorFlow, and large language models, with a strong foundation in MLOps, DevOps, CI/CD, and infrastructure as code using Terraform. Experienced designing RESTful APIs and scalable ML pipelines that integrate AI capabilities into production systems, including agentic AI applications. Known for balancing rapid innovation with reliability, security, and compliance in regulated environments.

SKILLS

- Machine Learning & AI: Large Language Models (LLMs) · Agentic AI Applications · Supervised & Unsupervised Learning · Deep Learning · NLP · Transformers · Model Training & Deployment · Model Optimization
- Programming: Python · Pandas · NumPy · PyTorch · TensorFlow · SQL · TypeScript
- Cloud & Infrastructure: AWS · Azure · Google Cloud Platform · Terraform (IaC) · Kubernetes · Docker · NVIDIA Triton
- MLOps & DevOps: CI/CD (GitHub Actions, Azure DevOps, Jenkins) · GitHub · ML Pipelines · Model Versioning · Automated Testing · Observability · Infrastructure Automation
- Backend & APIs: RESTful APIs · Microservices · FastAPI · gRPC · Distributed Systems
- Data Engineering: ETL Pipelines · Feature Engineering · Data Warehousing · BigQuery · Apache Beam · Dataflow
- Collaboration: Code Reviews · Cross-functional Collaboration · Technical Leadership · Agile/Scrum · Stakeholder Communication

PROFESSIONAL EXPERIENCE

Machine Learning Engineer | Meta | Dallas, TX | Jul 2022 – Present

- Designed, trained, and deployed large-scale deep learning and transformer-based models (PyTorch, TensorFlow) supporting conversational AI and multimodal systems used across Meta's platforms.
- Built agentic AI prototypes that combined LLM reasoning with tool use (search, retrieval, and structured APIs), enabling autonomous task execution and contextual decision-making workflows.
- Architected end-to-end ML pipelines in Python using Pandas and NumPy for preprocessing, feature engineering, training, evaluation, and deployment.
- Developed RESTful APIs (FastAPI) to expose model inference services, enabling seamless integration with product engineering teams across multiple applications.
- Partnered with AI platform teams to productionize models on Kubernetes using Docker and NVIDIA Triton, ensuring scalable, low-latency inference across GPU clusters.
- Implemented CI/CD workflows in GitHub Actions to automate testing, model validation, container builds, and deployment to cloud environments (AWS and GCP).
- Used Terraform to provision cloud infrastructure, including compute clusters, storage, and networking resources, aligning with infrastructure-as-code best practices.
- Contributed to code reviews, advocated for robust design patterns, and balanced rapid experimentation with reliability, security, and compliance requirements.
- Mentored engineers on MLOps, version control (GitHub), and best practices for building maintainable ML systems in highly distributed environments.

Technologies & Skills:

Python · Pandas · NumPy · PyTorch · TensorFlow · LLMs · Agentic AI · FastAPI · RESTful APIs · Kubernetes · Docker · NVIDIA Triton · AWS · GCP · Terraform · GitHub · CI/CD · ML Pipelines · Distributed Training · MLOps

Machine Learning Engineer | The Michaels Companies, Inc | Irving, TX | Aug 2020 – Jul 2022

- Designed and implemented machine learning algorithms for recommendation systems and customer segmentation used across e-commerce and in-store platforms.

- Built scalable ETL and feature engineering pipelines using Python, Pandas, SQL, Apache Beam, and BigQuery to support model training and real-time inference.
- Deployed deep learning and gradient-boosting models (PyTorch, TensorFlow, scikit-learn) to production using Docker and Kubernetes, enabling personalized product recommendations for millions of users.
- Developed RESTful microservices to integrate AI models directly into web and mobile applications, collaborating closely with line-of-business engineering teams.
- Established CI/CD pipelines in GitHub and Azure DevOps for automated model testing, containerization, and cloud deployment.
- Migrated ML workloads to Google Cloud Platform and Azure, provisioning infrastructure with Terraform to improve scalability and reduce operational overhead.
- Implemented monitoring and observability for model performance and data drift, ensuring system reliability in a retail environment handling sensitive customer data.
- Worked cross-functionally with product, analytics, and engineering teams in Agile sprints, conducting code reviews and clearly communicating technical trade-offs to non-technical stakeholders.

Technologies & Skills:

Python · Pandas · NumPy · PyTorch · TensorFlow · scikit-learn · SQL · BigQuery · Apache Beam · GCP · Azure · Terraform · Docker · Kubernetes · RESTful APIs · GitHub · CI/CD · ETL Pipelines · MLOps

Data Scientist | University of Maryland, Baltimore County | Baltimore, MD | Feb 2018 – May 2020

- Designed and trained supervised and unsupervised machine learning models using Python, scikit-learn, TensorFlow, Pandas, and NumPy to analyze large academic and operational datasets.
- Built reproducible ML pipelines for preprocessing, model evaluation, and hyperparameter tuning, improving research reliability and repeatability.
- Developed data warehousing and SQL-based data integration processes to support analytics and predictive modeling initiatives.
- Collaborated with research and IT teams to deploy analytical tools via internal web services and dashboards.
- Presented findings to technical and non-technical stakeholders, translating complex ML concepts into actionable insights.

Technologies & Skills:

Python · Pandas · NumPy · scikit-learn · TensorFlow · SQL · ML Pipelines · Data Warehousing · Model Evaluation · Research Communication

Research Assistant | University of Maryland, Baltimore County | Baltimore, MD | Sep 2016 – Feb 2018

- Assisted in developing predictive models for cybersecurity anomaly detection using supervised learning and NLP techniques.
- Implemented data preprocessing and feature extraction pipelines in Python, leveraging Pandas and NumPy for large-scale datasets.
- Experimented with early deep learning architectures and contributed to performance benchmarking and documentation.
- Used Git for version control and collaborated with faculty and graduate researchers in structured code reviews.

Technologies & Skills:

Python · Pandas · NumPy · scikit-learn · TensorFlow · SQL · Git · NLP · Data Preprocessing · Model Benchmarking

EDUCATION

Master's degree in Computer Science

University of Maryland – College Park, MD | 2018 – 2020

Bachelor's degree in Computer Science

University of Maryland, Baltimore County – Baltimore, MD | 2013 – 2017